Lesson 3: Take Your Shot

Scratch

VIDEO GAME CREATION

This week: spawning objects at runtime. Your job is to make your player shoot a projectile that flies up the screen and cleans itself up when it leaves - the first piece of your capstone's actual shooting mechanic.

Open your own [YourName]Game project from Lesson 2 - don't start a new one. Everything this week builds on the player object you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder projectile image (bullet_yellow.png) and a new sound effect (sfx_shoot.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 2.
2. Click the Sprite pane and create a new sprite for the projectile - import bullet_yellow.png from this lesson's Assets folder (Upload Sprite), or draw a small shape of your own.
3. Rename this new sprite Bullet, then hide it in the initial state (it should only appear as a clone, never as itself) - add a "when green flag clicked" script to the Bullet sprite with just a "hide" block.

Level 1 - Minimal (MVP)

Pressing the fire key spawns one projectile that flies upward and disappears when it leaves the screen. This is a complete finish for today - reaching this is enough to move on to next week.

4. On the Player sprite, click Events, drag "when [space] key pressed" into a new script.
5. Click the Control category, drag "create clone of [Bullet]" underneath it - this is a separate, independent script from your Player's movement scripts.
6. Switch to the Bullet sprite. Click Control, drag "when I start as a clone" into a new script area.
7. Under it, add "show", then a Motion "go to x: (Player's x) y: (Player's y)" block (Sensing > "[Player]'s [x position]" reporters dragged into the x/y slots).
8. Add a "forever" loop (Control) containing "change y by 10" (Motion).
9. Inside that same forever loop, add "if <y position > 170> then delete this clone" (Control > Sensing/Operators > Motion).
10. Click the stage, then press spacebar to test - a bullet should appear at the Player and fly upward, then vanish at the top.

Level 2 - Complete (Alpha/Beta)

Firing repeatedly produces multiple independent projectiles on screen at once, each moving and disappearing correctly, and rapid-fire is capped so holding the key down doesn't flood the screen.

- Test firing several times quickly - each press should produce its own independent bullet, all moving and disappearing on their own. Clones already behave this way automatically in Scratch, since each clone runs its own full copy of the Bullet's scripts.
- Add a cooldown so holding spacebar doesn't fire a new bullet every single frame: on the Player, add a "can_shoot" variable, set it to true at the start, and only "create clone of [Bullet]" when can_shoot is true - immediately set it to false, then use a short "wait 0.2 seconds" before setting it back to true.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: play sfx_shoot.wav (upload it from this lesson's Assets folder) the moment a bullet is fired - add "start sound [sfx_shoot]" right next to "create clone of [Bullet]" on the Player's fire script.
- Visual: give the Bullet its own look distinct from the Player - a different costume, size, or a brief "change size by" pulse when it spawns, so it reads clearly as "a shot" rather than a copy of the character.
- Structure: keep every piece of a clone's own behavior - appearing, moving, and cleaning itself up - inside that one "when I start as a clone" script on the Bullet sprite, and keep the Player's fire script limited to just triggering it (create clone + cooldown + sound). Don't let Bullet logic leak into the Player's scripts, or vice versa.

If Something's Wrong

- Nothing happens on spacebar: make sure "create clone of" is targeting Bullet (not "myself"), and that the Bullet sprite's "when I start as a clone" script actually has "show" - clones inherit whatever visibility the original had, which is hidden here.
- Project freezes or becomes unresponsive: this almost always means a clone is creating another clone inside its own "when I start as a clone" script - the fire logic belongs on the Player, not the Bullet.
- Bullets never disappear: double check the y > 170 comparison is inside the forever loop (not after it), and that the stage's visible area is roughly -180 to 180 for y.
- Holding a direction key, then pressing spacebar to fire, makes movement pause and the held arrow key stop responding: this is a known Scratch quirk, not a bug in your project - "when key pressed" scripts repeat by riding your operating system's own keyboard auto-repeat, and most operating systems only actively auto-repeat one key at a time, so a second key press can interrupt the first key's repeat stream. Release and press the movement key again to resume. See this lesson's Sidebars for why, and the Super-Stretch Ideas list for a way to rebuild movement so this can't happen.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas